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FEATURES OF PIERCING MILL TPA 50-200 WORKING ROLL WEAR DURING ROLLING CONTINUOUSLY-CAST AND HOT-ROLLED BILLETS

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Results are provided for a study of the effect of barrel-shaped and bowl-shaped working rollers of a screw rolling piercing mill with guide bars of a 50–200 pipe rolling unit. By using a fundamentally new concept for describing the wear mechanism of a roller working surface, a dependence is revealed for worn part volume on amount of rolled continuously-cast and hot-rolled billets 90–260 mm in diameter. The level of material wear durability and working roller loss, restored by hard facing with flux-cored wire grade 18KhGSA in relation to amount of pierced billets, are established.

Keywords: *wear, working rollers, material life, continuously-cast billet, hot-rolled billet, piercing, screw rolling mill.*

Extensive attention is paid in scientific publications to questions of wear resistance of a screw rolling machine tool, predominantly the piercing mill mandrel [1, 2], and also prediction of the behavior of metal at the site of deformation and loading on a working tool [3, 4]. However, questions of wear resistance of screw rolling mill working rollers and the effect of wear on rolling stability, bush and pipe quality, have not been studied adequately.

Numerous researchers [5, 6], including by modeling piercing if screw rolling mills [7], and also the practice pipe rolling manufacture [6], have established that bush and pipe quality with respect to internal surface condition depends to a considerable extent on the amount and quality of individual reductions, and ovalization of a billet at the deformation site of a piercing mill. All these parameters change significantly with wear of the roller working surface.

In fact, the number of individual reductions before the mandrel nose and pinching of rollers is sharply reduced, and also there is redistribution of ovalization over the site length. With increasing roller wear, some increase in observed in billet-sleeve ovality, caused mainly by a reduction in distance between rollers in order to compensate for wear, with almost unchanged or increasing distance between guides. It is natural that with an increase in roller wear there is a change in many other indices of the piercing process, i.e., worsening of conditions of gripping a billet by rollers, an increase in metal sliding at a deformation site, which lead to disruption of rolling, a reduction in mill productivity, and in a number of cases to worse finished pipe quality. Therefore, it is possible to use rollers to a certain extent up to a specific level of wear.

The authors in [8] studied in detail the effect of wear of barrel-shaped working rolls of steel 50X of different versions of sizing on the number of rolled continuously-cast billets of considerable diameter into sleeves in a mill with linear guides and description of the working roller wear mechanism was proposed.

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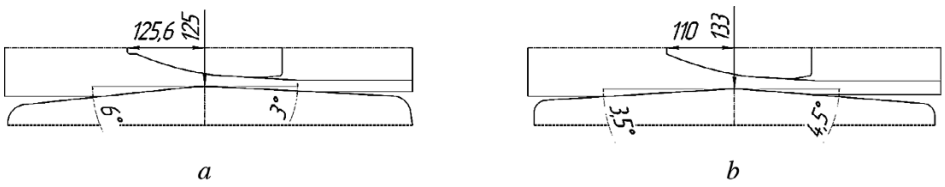


Fig. 1. Diagram of deformation site with piercing of a billet 156 mm in diameter using rollers: *a*) bowl-shaped (sleeve 142.5 × 19 mm); *b*) barrel-shaped (sleeve 151 × 22.5 mm).

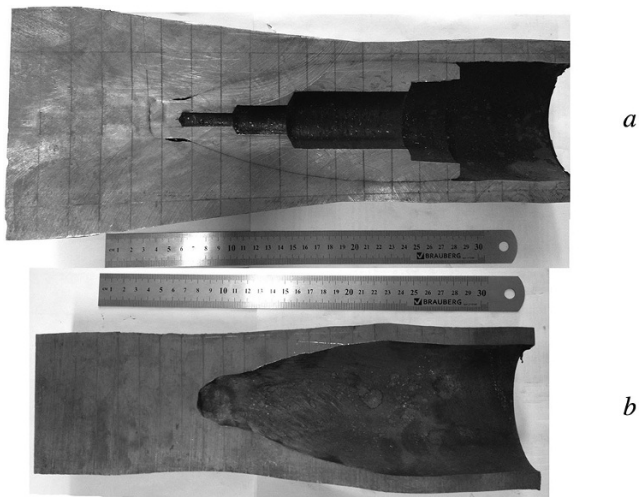


Fig. 2. External appearance of billet-sleeve, forcibly stopped during rolling in a piercing mill using bowl-shaped (*a*) and barrel-shaped (*b*) rollers.

Working roller material life is short and depends on sizing and adjustment of a piercing mill [6], which makes it possible to analyze features of roller wear for piercing continuously-cast and hot-rolled billets of carbon and high-carbon steels 90–260 mm in diameter and 1.4–4.3 m long, used for preparing hot-rolled pipes in a pipe-rolling unit TPA 50-200 at the Volga Pipe Plant. A feature of TPA 50-200 is use of billets of a broad range of diameters, in contrast to TPA with a continuous mill of the type MPM [8], PQF [9], or FGM [10] for which the range of diameters is limited to two or three standard sizes.

Billet piercing in a TPA 50-200 line is performed in a two-roller screw rolling mill using barrel-shaped or bowl-shaped rollers arranged in a horizontal plane and directional guides. The maximum barrel of the working rollers is 1100 mm. Restoration of the working surface of steel 45 working rollers is accomplished by regrinding with a reduction in diameter by a value up to 14%, and number of regrindings from four to eight. New rollers have hardness of 190–210 HB, although as they are reground, the working surface hardness decreases to 145–157 HB. Hard facing is also used for working roller restoration with flux-cored wire grade 18KhGSA. Compared with steel 45 rollers, the roller surface has greater hardness (230–245 HB), and, as they are reground, the working surface hardness is reduced to 150–170 HB.

On the basis of practical operation of TPA 50-200, it has been established that for stable performance of piercing billets of carbon steels the amount of reduction used ahead of the nose of a mandrel is up to 7.0%, and for highly alloyed steels, it is 4.0–5.0%. Billet reduction in pinching of bowl-shaped rollers is 11–31% [11], and for barrel-shaped rollers, it is 9–16%, with feed angles of 10–14° (Fig. 1). The ovalization coefficient is within the limits 1.09–1.11.

Piercing of billets using bowl-shaped rollers is carried out in preparing sleeves of small diameter compared with a billet (Fig. 2) up to 24% [11]. The use of bowl-shaped rollers is implemented successfully in the TPA 50-200 in piercing of carbon and alloy steels, including bearing steels.

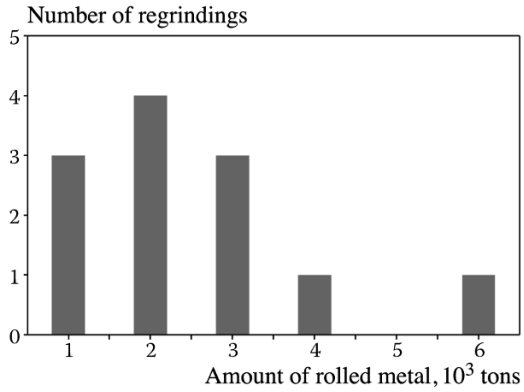


Fig. 3. Dependence of guide working roller regrinding on amount of rolled metal.

TABLE 1. Piercing Mill Working Roller Calibration Parameters

Working roll type	Diameter, mm		Feed angle, deg	Deformation site cone generating line slope, deg	
	billet	sleeve		entry φ_1	exit φ_2
Bowl-shaped	150–230	114–223	13–14	6.0	3.0
Barrel-shaped	90–260	91–259	10–12	3.5	4.5

Analysis of values of drawing coefficient μ in preparing sleeves of a large range of TPA with a ratio of diameter to wall thickness of 5.5–6.7 has made it possible to establish that in the majority of cases billet piercing in bowl-shaped rollers is accomplished with $\mu = 2.6$; in barrel-shaped rollers, $\mu = 1.8$.

Analysis of statistical data for the period of February to April 2016 for removal of twelve pairs of rollers form operation (for planned repair due to surface working defects, reaching the barrel diameter) makes it possible to conclude that often the working rollers of a TPA 50-200 piercing mill are changed and reground to a smaller diameter in view of reaching a controlled number of rolled billets.

Most often roller regrinding occurs on reaching an operating time of rolling up to 2000 tons of rolled metal (Fig. 3). However, three pairs of rollers had running time from regrinding to regrinding in the range up to 1000 tons, and one pair of rollers had running time from 5000 to 6000 tons of rolled metal.

During research, a study of bowl-shaped and barrel-shaped working roller wear was made (Table 1).

The depth of working roller wear in the piercing cone and billet were determined by means of a gauge marked with a pitch of 15 mm and feelers with thickness up to 0.1 mm. In addition, working surface hardness was monitored with an interval of 15 mm over the length of a roller barrel (Figs. 4 and 5).

It was established that as in other piercing mills [5, 6, 8] the most intense wear occurs in the area of gripping a billet by the working rollers, mainly, as a result of mechanical damage of a roller by the sharp edges of the end of a billet not rotating at the instant of feeding into the mill. In view of this, an understanding of "roller wear" means the depth of wear of the entry cone that changes the original sizing of rollers.

It should be noted that billets 240–260 mm in diameter with a maximum length of 4.4 m and weight up to 1.8 tons are coated on the outer side with a layer of furnace scale, which falls off in to a piercing mill receiving trough, prevents rotation of a billet at the instant of gripping in working rollers, and causes sliding of a billet with respect to rollers and its polishing.

Rollers wear especially rapidly in the initial operating period. Roller wear in this period in the entry cone occurs almost uniformly over the length, and gripping of a billet due to presence of burnishing in the entry cone to a depth of 0.5 mm

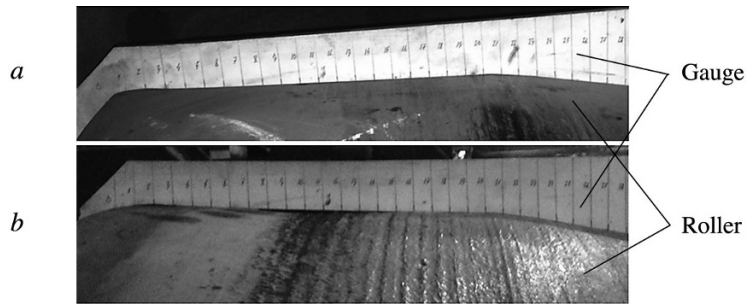


Fig. 4. External appearance of inlet cone of worn bowl-shaped rollers: *a*) left-hand roller after rolling 723 tons; *b*) right-hand roller after rolling 1529 tons.



Fig. 5. External appearance of inlet cone of worn barrel-shaped rollers: *a*) left-hand roller after rolling 196 tons; *b*) right-hand roller after rolling 5818 tons.

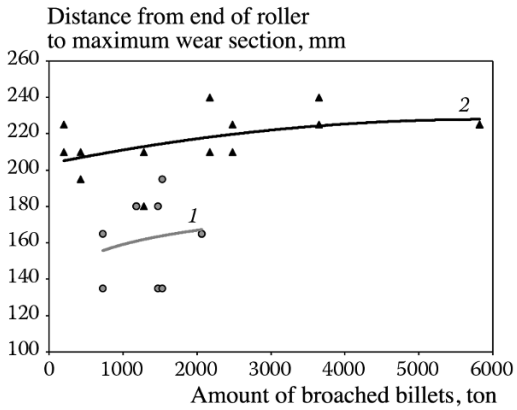


Fig. 6. Dependence of distance from end of working roller to maximum wear section on amount of broached billets: *1*) bowl-shaped rollers; *2*) barrel-shaped rollers.

is accomplished stably. A gradual increase in depth of maximum wear slows down. The section of maximum wear h_{max} of working rollers moves in the rolling direction. For example, during piercing of 723 tons of billets of different diameters in bowl-shaped rollers, h_{max} is at distance of 135 mm from the end of a roller, and after piercing 2062 tons, it is at a distance of 165 mm; with piercing of 196 tons of billets of different diameters in barrel-shaped rollers, h_{max} is at a distance 210 mm from the end of a roller, but after piercing 5818 tons, it is at a distance of 225 mm (Fig. 6).

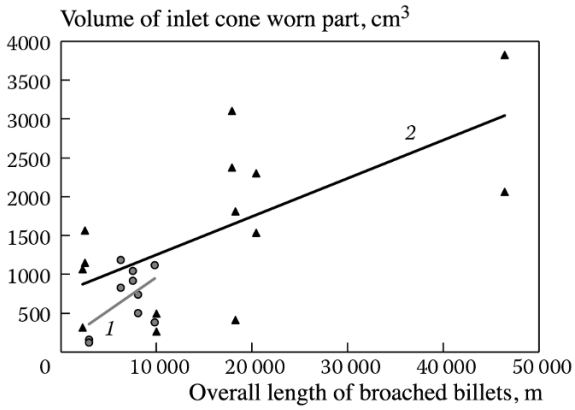


Fig. 7. Dependence of working roller entry cone worn part volume on overall length of broached billets: *1*) bowl-shaped rollers; 2) barrel-shaped rollers.

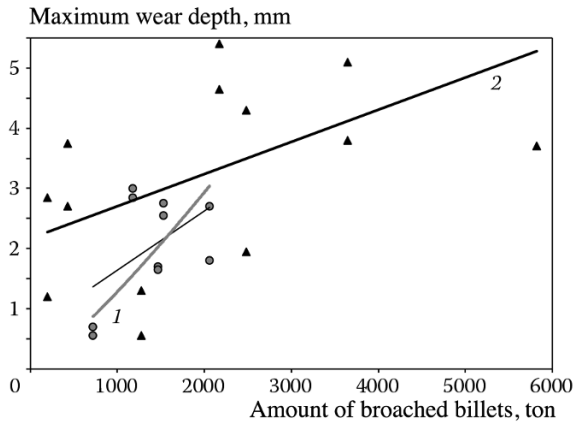


Fig. 8. Effect of roller sizing on nature of wear: *1*) bowl-shaped rollers; *2*) barrel-shaped rollers.

With the aim of determining the amount of wear of the inlet cone, i.e, the amount of roller material expended in gripping a billet, in relation to the amount of pierced billets, a procedure given in [8] was used. During research, the dependence of volume of wear of the inlet cone of working rollers on overall length of pierced billets in the TPA 50-200 was also analyzed. As is seen from the data provided in Fig. 7, there is an almost linear relationship between the volume of worn part of a roller and overall length of pierced billets, as also for the amount of rolled metal [6, 8].

An analysis of the data for wear of 12 pairs of bowl-shaped and barrel-shaped rollers of two sizes showed that the material durability of hard-faced rollers of a screw rolling piercing mill TPA 50-200 is 0.17–5.52 tons/cm^3, and the average value is 2.1 tons/cm^3, which is significantly lower than the material durability of rollers of piercing mills TPA 30-102 [6] and TPA 159-426 [8]. Apparently, this is due to the effect of a considerable amount of standard size of pierced billets and large inlet cone angle (6°) in bowl-shaped rollers.

Proceeding from analysis of a considerable volume of statistical data, it may be concluded that the depth of maximum wear for bowl-shaped rollers in the initial operating period is less by a factor of 2.5 than the depth of maximum wear for barrel-shaped rollers (Fig. 8). However, as the amount of pierced billets increases, the intensity of increase in depth of maximum wear for bowl-shaped rollers is significantly greater than for barrel-shaped rollers.

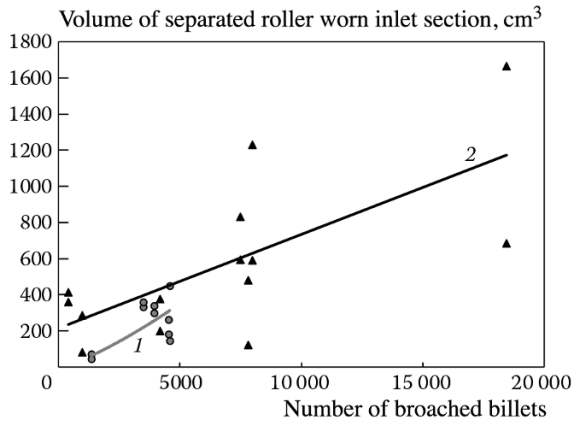


Fig. 9. Dependence of separated worn volume of roller entry section on amount of broached billets: *1*) bowl-shaped; *2*) barrel-shaped rollers.

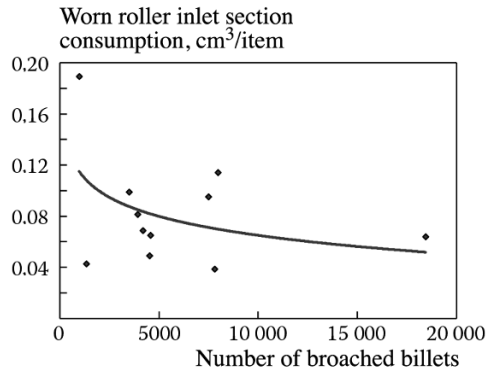


Fig. 10. Dependence of roller inlet cone material consumption on amount of broached billets.

Using a dependence for inlet cone wear index as a result of abrasive wear expended in gripping billets [8], from the volume of data collected for wear of inlet and outlet parts of working rollers of a piercing mill, a curve has been plotted for the dependence of volume of the worn inlet part of a roller on amount of pierced billets (Fig. 9). As is seen from the curve given in Fig. 9, there is a linear connection between the volume of the worn part of the inlet cone of rollers as a result of wear and the amount of pierced billets.

For the sizing versions in question, the ratio of worn part of a roller to volume of worn part of the inlet cone expended in gripping billets, depending on amount, comprised: 3.0 for bowl-shaped rollers and 3.4 for barrel-shaped rollers

A dependence is given in Fig. 10 for consumption of material of the worn entry part of right-hand and left-hand bowl-shaped and barrel-shaped rollers on amount of rolled billets. It is apparent that this relationship has a tendency towards a continuous reduction, and this may be explained by strengthening of roller material in the area of maximum wear. Values of the worn inlet part of a roller consumption vary from 0.19 cm^3/item with piercing of 969 billets to 0.06 cm^3/item with piercing of 18439 billets. The average value of consumption as a result of abrasive wear comprises 0.08 cm^3 for one pieced billet.

It should be noted that as there is an increase in amount of rolled billets there is an increase in hardness in the area of maximum roller wear (Fig. 11), and there is an increase in material life.

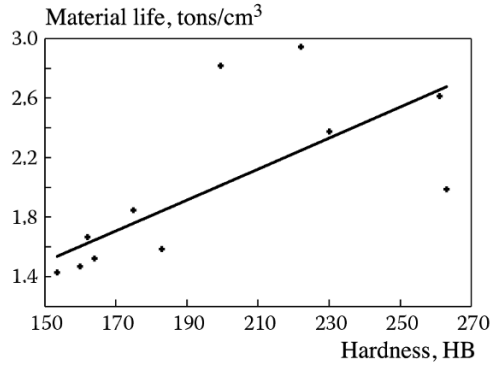


Fig. 11. Effect of roller working surface hardness in maximum wear area on material life.

As a result of studying wear of the working surface of screw Volga Pipe Plant TPA 50-200 rolling piercing mill rollers, the following have been established:

1) the average values for material durability for bowl-shaped and barrel-shaped rollers is 2.1 tons/cm^3, and from 196 to 5818 tons of billets were rolled before roller regrinding;

2) during operation, the section of maximum wear for bowl-shaped and barrel-shaped rollers moves in the rolling direction, although the rate of this movement decreases continuously;

3) the depth of maximum wear of bowl-shaped rollers in the initial operating period is less by a factor of 2.5 than the depth of maximum wear for barrel-shaped rollers. As the amount of billet rolled metal increases, the intensity of increase in maximum wear of bowl-shaped rollers is greater than for barrel-shaped rollers; and

4) the dependence of consumption of the worn inlet part of roller on amount of pierced billet (see above) has a tendency towards a steady reduction. The average amount of material consumed for the roller inlet cone as a result of abrasive wear is 0.08 cm^3 for one pierced billet.

Conclusion. The wear mechanism has been studied for bowl-shaped and barrel-shaped working rollers of screw rolling piercing mill TPA 50-200 with piercing of billets form 90 to 260 mm in diameter.

An abrasive wear index has been derived for the inlet cone in relation to amount of rolled metal, which is 0.08 cm^3 for one pierced billet.

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